

Sampling Distributions & the Central Limit Theorem



What is a sampling distribution?

- 1 Explain the difference between a population distribution and the sampling distribution of a sample mean $\bar{X}$.
 - 2 Explain why the sampling distribution of $\bar{X}$ has less variability than the original population distribution.
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Mean and standard deviation of the sampling distribution

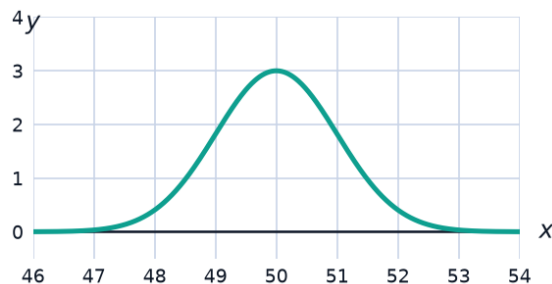
- 3 A population has mean $\mu = 60$ and standard deviation $\sigma = 12$. For samples of size $n = 36$, find the mean and standard deviation of the sampling distribution of $\bar{X}$.
 - 4 Using the same population, find the standard error for a larger sample of size $n = 144$, and explain what happens to the standard error as sample size increases.
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The Central Limit Theorem

- 5 State the Central Limit Theorem in your own words, including what happens to the shape of the sampling distribution of $\bar{X}$ as sample size n increases.
 - 6 Explain why the Central Limit Theorem is considered one of the most important results in statistics, referencing its practical usefulness.
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Applying the Central Limit Theorem

- 7 A population of package weights has mean $\mu = 50$ lbs and standard deviation $\sigma = 8$ lbs (population shape unknown). For a random sample of $n = 64$ packages, find the probability the sample mean weight exceeds 52 lbs, given $P(Z > 2) \approx 0.0228$.



Sampling distribution of a sample proportion

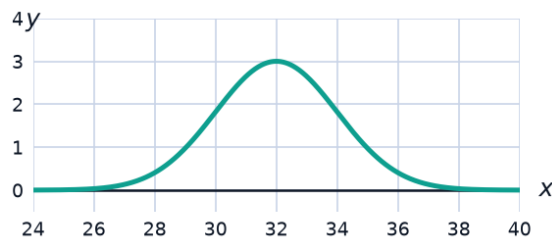
- 8 A population has a true proportion $p = 0.4$ favoring a proposal. For samples of size $n = 100$, find the mean and standard deviation of the sampling distribution of the sample proportion $\hat{p}$, using $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$.
- 9 State the two conditions typically required to treat the sampling distribution of $\hat{p}$ as approximately normal.
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Comparing sampling variability across sample sizes

- 10 Explain, in the context of opinion polling, why a poll of 1000 people gives a more reliable estimate of a population proportion than a poll of 50 people, connecting your answer to the formula for $\sigma_{\hat{p}}$.
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Applying the sampling distribution of the sample mean

- 11 A population of commute times has mean $\mu = 32$ minutes and standard deviation $\sigma = 10$ minutes. For a random sample of $n = 25$ commuters, find the probability the sample mean commute time exceeds 35 minutes, given $P(Z > 1.5) \approx 0.0668$.



Sample size and the CLT threshold

- 12 A population of insurance claim amounts is strongly right-skewed. Explain why a sample of size $n = 15$ might not be large enough to safely use CLT-based normal methods, but a sample of size $n = 200$ likely would be.
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Standard error vs. standard deviation

- 13 Explain the difference between the standard deviation of an individual population value and the standard error of a sample mean $\bar{x}$.

- 14 A quality control engineer at a bottling plant knows that individual bottle fill volumes are approximately normal with mean 500 mL and standard deviation 8 mL. Every hour, the engineer measures a sample of 16 bottles and computes the sample mean fill volume. Describe the shape, mean, and standard deviation of the sampling distribution of this hourly sample mean, and explain whether the Central Limit Theorem's usual $n \geq 30$ guideline is even needed to justify treating this sampling distribution as normal.
- 15 A political pollster samples 900 voters and calculates a sample proportion $\hat{p} = 0.53$ supporting a ballot measure, from a true population proportion of $p = 0.50$. Verify that both conditions ($np \geq 10$ and $n(1 - p) \geq 10$) are satisfied for treating the sampling distribution of $\hat{p}$ as approximately normal, then find the standard deviation of this sampling distribution, to four decimal places.
- 16 A large factory produces light bulbs whose individual lifespans are strongly right-skewed (most bulbs fail early, but a few last an unusually long time), with population mean $\mu = 800$ hours and standard deviation $\sigma = 250$ hours. An engineer takes a random sample of $n = 64$ bulbs and computes the sample mean lifespan. Explain, step by step, why it is reasonable to treat this sampling distribution of $\bar{x}$ as approximately normal despite the population itself being skewed, then compute the standard error of $\bar{x}$.
- 17 An online retailer knows that 22% of all website visitors make a purchase. On a particular day, the retailer records the browsing sessions of a random sample of 500 visitors and calculates the sample proportion who made a purchase. Verify the two conditions needed to treat this sampling distribution as approximately normal, then compute the standard deviation of the sampling distribution of the sample proportion, to four decimal places, and describe what this standard deviation represents in the context of comparing many different days' worth of visitor samples.

Multiple choice — easy 15

- 18 A population has mean $\mu = 100$ and standard deviation $\sigma = 60$. For samples of size $n = 100$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error), using $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.
- a. 6
 - b. 5
 - c. 7
 - d. 60

- 19 A population has mean $\mu = 100$ and standard deviation $\sigma = 200$. For samples of size $n = 100$, find the standard deviation of the sampling distribution of $\bar{X}$ (the standard error), using $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.
- a. 21
 - b. 19
 - c. 20
 - d. 200
- 20 A population has mean $\mu = 100$ and standard deviation $\sigma = 117$. For samples of size $n = 169$, find the standard deviation of the sampling distribution of $\bar{X}$ (the standard error), using $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.
- a. 10
 - b. 117
 - c. 9
 - d. 8
- 21 A population has mean $\mu = 100$ and standard deviation $\sigma = 88$. For samples of size $n = 121$, find the standard deviation of the sampling distribution of $\bar{X}$ (the standard error), using $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.
- a. 8
 - b. 88
 - c. 7
 - d. 9
- 22 A population has mean $\mu = 100$ and standard deviation $\sigma = 52$. For samples of size $n = 169$, find the standard deviation of the sampling distribution of $\bar{X}$ (the standard error), using $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.
- a. 4
 - b. 5
 - c. 3
 - d. 52

- 23 A population has mean $\mu = 100$ and standard deviation $\sigma = 90$. For samples of size $n = 324$, find the standard deviation of the sampling distribution of $\bar{X}$ (the standard error), using $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.
- a. 90
 - b. 4
 - c. 6
 - d. 5
- 24 A population has mean $\mu = 100$ and standard deviation $\sigma = 21$. For samples of size $n = 49$, find the standard deviation of the sampling distribution of $\bar{X}$ (the standard error), using $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.
- a. 2
 - b. 3
 - c. 4
 - d. 21
- 25 A population has mean $\mu = 100$ and standard deviation $\sigma = 26$. For samples of size $n = 4$, find the standard deviation of the sampling distribution of $\bar{X}$ (the standard error), using $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.
- a. 12
 - b. 13
 - c. 26
 - d. 14
- 26 A population has mean $\mu = 100$ and standard deviation $\sigma = 340$. For samples of size $n = 289$, find the standard deviation of the sampling distribution of $\bar{X}$ (the standard error), using $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.
- a. 340
 - b. 19
 - c. 20
 - d. 21

- 27 A population has mean $\mu = 100$ and standard deviation $\sigma = 84$. For samples of size $n = 36$, find the standard deviation of the sampling distribution of $\bar{X}$ (the standard error), using $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.
- a. 14
 - b. 13
 - c. 15
 - d. 84
- 28 A population has mean $\mu = 100$ and standard deviation $\sigma = 288$. For samples of size $n = 256$, find the standard deviation of the sampling distribution of $\bar{X}$ (the standard error), using $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.
- a. 17
 - b. 19
 - c. 18
 - d. 288
- 29 A population has mean $\mu = 100$ and standard deviation $\sigma = 140$. For samples of size $n = 49$, find the standard deviation of the sampling distribution of $\bar{X}$ (the standard error), using $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.
- a. 140
 - b. 20
 - c. 21
 - d. 19
- 30 A population has mean $\mu = 100$ and standard deviation $\sigma = 72$. For samples of size $n = 16$, find the standard deviation of the sampling distribution of $\bar{X}$ (the standard error), using $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.
- a. 72
 - b. 19
 - c. 17
 - d. 18

- 31 A population has mean $\mu = 100$ and standard deviation $\sigma = 49$. For samples of size $n = 49$, find the standard deviation of the sampling distribution of $\bar{X}$ (the standard error), using $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.
- a. 49
 - b. 8
 - c. 7
 - d. 6
- 32 A population has mean $\mu = 100$ and standard deviation $\sigma = 99$. For samples of size $n = 121$, find the standard deviation of the sampling distribution of $\bar{X}$ (the standard error), using $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.
- a. 8
 - b. 99
 - c. 10
 - d. 9

Multiple choice — medium 15

- 33 A population has a true proportion $p = 0.40$. For samples of size $n = 200$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1 - p)/n}$.
- a. 0.0346
 - b. 0.0246
 - c. 0.2400
 - d. 0.0446

- 34 A population has a true proportion $p = 0.10$. For samples of size $n = 800$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1 - p)/n}$.
- a. 0.0206
 - b. 0.0900
 - c. 0.0006
 - d. 0.0106
- 35 A population has a true proportion $p = 0.10$. For samples of size $n = 300$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1 - p)/n}$.
- a. 0.0273
 - b. 0.0900
 - c. 0.0173
 - d. 0.0073
- 36 A population has a true proportion $p = 0.80$. For samples of size $n = 50$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1 - p)/n}$.
- a. 0.0566
 - b. 0.1600
 - c. 0.0466
 - d. 0.0666
- 37 A population has a true proportion $p = 0.40$. For samples of size $n = 400$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1 - p)/n}$.
- a. 0.0345
 - b. 0.2400
 - c. 0.0245
 - d. 0.0145

- 38 A population has a true proportion $p = 0.70$. For samples of size $n = 300$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1 - p)/n}$.
- a. 0.0365
 - b. 0.0265
 - c. 0.0165
 - d. 0.2100
- 39 A population has a true proportion $p = 0.90$. For samples of size $n = 300$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1 - p)/n}$.
- a. 0.0073
 - b. 0.0173
 - c. 0.0273
 - d. 0.0900
- 40 A population has a true proportion $p = 0.10$. For samples of size $n = 50$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1 - p)/n}$.
- a. 0.0324
 - b. 0.0524
 - c. 0.0424
 - d. 0.0900
- 41 A population has a true proportion $p = 0.60$. For samples of size $n = 200$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1 - p)/n}$.
- a. 0.0346
 - b. 0.0446
 - c. 0.0246
 - d. 0.2400

- 42 A population has a true proportion $p = 0.60$. For samples of size $n = 50$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1 - p)/n}$.
- a. 0.2400
 - b. 0.0793
 - c. 0.0593
 - d. 0.0693
- 43 A population has a true proportion $p = 0.30$. For samples of size $n = 300$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1 - p)/n}$.
- a. 0.2100
 - b. 0.0365
 - c. 0.0165
 - d. 0.0265
- 44 A population has a true proportion $p = 0.90$. For samples of size $n = 600$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1 - p)/n}$.
- a. 0.0900
 - b. 0.0222
 - c. 0.0022
 - d. 0.0122
- 45 A population has a true proportion $p = 0.30$. For samples of size $n = 500$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1 - p)/n}$.
- a. 0.0305
 - b. 0.0205
 - c. 0.2100
 - d. 0.0105

- 46 A population has a true proportion $p = 0.20$. For samples of size $n = 200$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1 - p)/n}$.
- a. 0.0383
 - b. 0.0183
 - c. 0.1600
 - d. 0.0283
- 47 A population has a true proportion $p = 0.20$. For samples of size $n = 50$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1 - p)/n}$.
- a. 0.0566
 - b. 0.1600
 - c. 0.0666
 - d. 0.0466
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Multiple choice — hard 15

- 48 A population has mean $\mu = 143$ and standard deviation $\sigma = 90$ (population shape unknown). For a random sample of $n = 36$, find the probability the sample mean is above 158, using the Central Limit Theorem.
- a. 11.87%
 - b. 15.87%
 - c. 84.13%
 - d. 19.87%

- 49 A population has mean $\mu = 112$ and standard deviation $\sigma = 165$ (population shape unknown). For a random sample of $n = 121$, find the probability the sample mean is above 127, using the Central Limit Theorem.
- a. 84.13%
 - b. 19.87%
 - c. 11.87%
 - d. 15.87%
- 50 A population has mean $\mu = 48$ and standard deviation $\sigma = 132$ (population shape unknown). For a random sample of $n = 144$, find the probability the sample mean is below 81, using the Central Limit Theorem.
- a. 96.87%
 - b. 95.87%
 - c. 0.13%
 - d. 99.87%
- 51 A population has mean $\mu = 67$ and standard deviation $\sigma = 36$ (population shape unknown). For a random sample of $n = 144$, find the probability the sample mean is above 73, using the Central Limit Theorem.
- a. 8.28%
 - b. 97.72%
 - c. 2.28%
 - d. 6.28%

- 52 A population has mean $\mu = 88$ and standard deviation $\sigma = 105$ (population shape unknown). For a random sample of $n = 49$, find the probability the sample mean is below 118, using the Central Limit Theorem.
- a. 97.72%
 - b. 2.28%
 - c. 94.72%
 - d. 93.72%
- 53 A pollster samples $n = 500$ voters, from a population with true proportion $p = 0.20$ supporting a measure. Verify that both conditions ($np \geq 10$ and $n(1 - p) \geq 10$) are satisfied, then find the standard deviation of the sampling distribution of $\hat{p}$, to four decimal places.
- a. 0.0259
 - b. 0.2000
 - c. 0.0179
 - d. 0.0099
- 54 A pollster samples $n = 1200$ voters, from a population with true proportion $p = 0.70$ supporting a measure. Verify that both conditions ($np \geq 10$ and $n(1 - p) \geq 10$) are satisfied, then find the standard deviation of the sampling distribution of $\hat{p}$, to four decimal places.
- a. 0.7000
 - b. 0.0052
 - c. 0.0132
 - d. 0.0212

- 55 A pollster samples $n = 1000$ voters, from a population with true proportion $p = 0.20$ supporting a measure. Verify that both conditions ($np \geq 10$ and $n(1 - p) \geq 10$) are satisfied, then find the standard deviation of the sampling distribution of $\hat{p}$, to four decimal places.
- a. 0.0046
 - b. 0.0206
 - c. 0.0126
 - d. 0.2000
- 56 A pollster samples $n = 800$ voters, from a population with true proportion $p = 0.60$ supporting a measure. Verify that both conditions ($np \geq 10$ and $n(1 - p) \geq 10$) are satisfied, then find the standard deviation of the sampling distribution of $\hat{p}$, to four decimal places.
- a. 0.0253
 - b. 0.6000
 - c. 0.0173
 - d. 0.0093
- 57 A pollster samples $n = 500$ voters, from a population with true proportion $p = 0.60$ supporting a measure. Verify that both conditions ($np \geq 10$ and $n(1 - p) \geq 10$) are satisfied, then find the standard deviation of the sampling distribution of $\hat{p}$, to four decimal places.
- a. 0.6000
 - b. 0.0139
 - c. 0.0299
 - d. 0.0219

- 58 A population has standard deviation $\sigma = 160$. Find the sample size n needed so that the standard error $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ equals 20.
- a. 72
 - b. 64
 - c. 56
 - d. 8
- 59 A population has standard deviation $\sigma = 60$. Find the sample size n needed so that the standard error $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ equals 5.
- a. 144
 - b. 132
 - c. 12
 - d. 156
- 60 A population has standard deviation $\sigma = 180$. Find the sample size n needed so that the standard error $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ equals 10.
- a. 342
 - b. 18
 - c. 306
 - d. 324
- 61 A population has standard deviation $\sigma = 152$. Find the sample size n needed so that the standard error $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ equals 8.
- a. 342
 - b. 19
 - c. 380
 - d. 361

- 62 A population has standard deviation $\sigma = 204$. Find the sample size n needed so that the standard error $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ equals 17.
- a. 132
 - b. 144
 - c. 12
 - d. 156